

P–SV–SH Unitarity: Energy Flux, Open Channels and the Kennett Recursion

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Status. The interface results of §8 are validated by two independent implementations that agree to 10^{-15} : the Aki–Richards closed forms in `kennett_layers.py` (Python) and displacement–traction continuity in the thesis energy-normalised eigenbasis, `InterfaceUnitarity.wl` (Mathematica, 50 digits). The comparison is run by `scripts/gate_interface_unitarity.py`. The periodic-plane statement of §5 is the acceptance criterion of the intra-plane energy-balance gate and is not re-derived here.

1 The scattering matrix at fixed horizontal slowness

A laterally homogeneous layer, or a stack of layers, preserves the horizontal slowness p . At each p and frequency ω it therefore acts as a linear map from incoming to outgoing plane waves (Kennett, 1983; Kennett and Kerry, 1979). Write $\mathbf{a}$ for the incoming amplitudes (downgoing above, upgoing below) and $\mathbf{b}$ for the outgoing ones (upgoing above, downgoing below). Then

$$\mathbf{b} = \mathbf{S} \mathbf{a}, \quad \mathbf{S} = \begin{pmatrix} \mathbf{R}_D & \mathbf{T}_U \\ \mathbf{T}_D & \mathbf{R}_U \end{pmatrix}. \quad (1)$$

In an isotropic medium the in-plane (P–SV) and anti-plane (SH) motions decouple (Aki and Richards, 2002, ch. 5), so $\mathbf{S}$ is block diagonal with two independent pieces.

- **P–SV**: channels $\{P, SV\}$ on each side. $\mathbf{R}_D, \mathbf{R}_U, \mathbf{T}_D, \mathbf{T}_U$ are 2×2 and $\mathbf{S}$ is 4×4 .
- **SH**: one channel on each side. The blocks are scalars and $\mathbf{S}$ is 2×2 .

The unitarity statement

$$\mathbf{S}^\dagger \mathbf{S} = \mathbf{I} \quad (2)$$

is therefore two statements, one for each block.

2 Why flux normalisation is required

Equation (2) expresses energy conservation only if $|a_k|^2$ is the vertical energy flux carried by channel k . For a plane wave of displacement amplitude A the time-averaged vertical flux is

$$\langle F_z \rangle = \frac{1}{2} \omega^2 \rho v^2 \eta_v |A|^2, \quad \eta_v = \sqrt{v^{-2} - p^2}, \quad (3)$$

with $v = \alpha$ for P and $v = \beta$ for SV and SH (Aki and Richards, 2002, ch. 5). For SH the prefactor is the vertical impedance $Z = \rho \beta^2 \eta_\beta$.

Displacement coefficients therefore obey an energy balance with channel weights, for example

$$1 = |R_{PP}|^2 + \frac{\rho_1 \beta_1^2 \eta_{\beta 1}}{\rho_1 \alpha_1^2 \eta_{\alpha 1}} |R_{SP}|^2 + \frac{\rho_2 \alpha_2^2 \eta_{\alpha 2}}{\rho_1 \alpha_1^2 \eta_{\alpha 1}} |T_{PP}|^2 + \frac{\rho_2 \beta_2^2 \eta_{\beta 2}}{\rho_1 \alpha_1^2 \eta_{\alpha 1}} |T_{SP}|^2. \quad (4)$$

Rescaling each amplitude by the square root of its flux weight removes the weights. The rescaled coefficients are the *energy-normalised* or *modified* coefficients (Frasier, 1970; Kennett and Kerry, 1979; Chapman, 2004). In these variables,

$$\sum_k |b_k|^2 = \sum_k |a_k|^2 \quad \text{for all } \mathbf{a} \iff \mathbf{S}^\dagger \mathbf{S} = \mathbf{I}. \quad (5)$$

The implementation uses $\sqrt{\rho \eta}$ factors for P-SV, with the velocity factors carried by Kennett's amplitude convention, and $\sqrt{Z}$ for SH.

SH example. At a welded interface the modified SH coefficients are

$$R = \frac{Z_1 - Z_2}{Z_1 + Z_2}, \quad T = \frac{2\sqrt{Z_1 Z_2}}{Z_1 + Z_2}, \quad \mathbf{S}_{\text{SH}} = \begin{pmatrix} R & T \\ T & -R \end{pmatrix}. \quad (6)$$

For real Z_1, Z_2 , $R^2 + T^2 = 1$ and $RT - TR = 0$, so $\mathbf{S}_{\text{SH}}$ is real orthogonal and hence unitary.

3 Content of the identity

Equation (2) says that the columns of $\mathbf{S}$ are orthonormal. The two parts constrain different things.

- **Unit column norms: energy.** For an incident downgoing P wave, $|R_{PP}|^2 + |R_{SP}|^2 + |T_{PP}|^2 + |T_{SP}|^2 = 1$. This is (4) with the weights absorbed.
- **Orthogonal columns: phase.** For distinct inputs $j \neq l$, $\sum_k S_{kj}^* S_{kl} = 0$. Two different incident waves produce outgoing fields with no net interference flux. A model can have correct energies and still fail this condition if its phases are wrong.

Table 1. Conditions under which (2) holds, and what happens when each one fails.

Condition	If violated
Lossless media (real α, β)	η is complex, $\sqrt{\rho\eta}$ is no longer a flux normalisation, and (2) carries no energy meaning. For a thick lossy layer propagation dominates and $\mathbf{S}^\dagger \mathbf{S} \leq \mathbf{I}$ approximately, the defect being absorbed energy.
Propagating channels only	Beyond a critical slowness, e.g. $p > 1/\alpha_2$, the channel is evanescent and carries no flux on its own. Remove it; the restriction of $\mathbf{S}$ to open channels is unitary.
No open diffraction orders (periodic plane)	Energy leaves through channels with $\mathbf{k}_\parallel + \mathbf{G} \neq \mathbf{k}_\parallel$. The specular $\mathbf{S}$ is sub-unitary even without loss.

4 Conditions for unitarity

Evanescent channels. An evanescent pair carries flux only through the interference of its up- and downgoing parts. If such a channel is kept, energy conservation is no longer $\mathbf{S}^\dagger \mathbf{S} = \mathbf{I}$ but a statement in a non-diagonal flux metric (Newton, 1982). Restricting to open channels avoids this. Near the critical slowness $\eta \rightarrow 0$, so the normalisation factor $\sqrt{\rho\eta}$ vanishes and the coefficients are poorly conditioned.

5 Periodic planes: why specular unitarity is legitimate

A plane of cubic voxels with lattice pitch a_L is laterally periodic, not laterally invariant. Horizontal wavenumber is conserved only modulo reciprocal-lattice vectors $\mathbf{G}$, and the unitary scattering matrix lives in one Bloch sector. It includes *every* open channel (mode, $\pm, \mathbf{G}$) with real vertical wavenumber

$$k_z(\mathbf{G}) = \sqrt{\kappa^2 - |\mathbf{k}_\parallel + \mathbf{G}|^2}, \quad \kappa \in \{\kappa_p, \kappa_s\} \quad (7)$$

(Strutt, 1907; Petit, 1980). When the plane is sub-wavelength, every $\mathbf{G} \neq 0$ order is evanescent. Only the specular order is open, and the 4×4 P-SV and 2×2 SH specular matrices are unitary by themselves. The plane then acts as an effective laterally homogeneous layer, which is the premise of an intra-plane Foldy–Lax reduction to $R/T(p)$. If an order opens, specular unitarity is false, and $\mathbf{I} - \mathbf{S}^\dagger \mathbf{S}$ measures diffracted energy rather than error. A numerical gate must refuse to run in that regime, not report a defect.

6 Unitarity versus reciprocity

These are independent properties.

- **Reciprocity** follows from the Betti–Rayleigh theorem. With Kennett’s phase convention for the modified coefficients it makes $\mathbf{S}$ symmetric: $\mathbf{S} = \mathbf{S}^\top$, i.e. $\mathbf{R}_D = \mathbf{R}_D^\top$, $\mathbf{R}_U = \mathbf{R}_U^\top$ and $\mathbf{T}_U = \mathbf{T}_D^\top$ (Kennett and Kerry, 1979; Varatharajulu and Pao, 1976). It is bilinear,

contains no complex conjugate, and *survives attenuation*. Its matrix form depends on the per-channel phase convention. In the thesis eigenbasis the P-SV block obeys instead

$$\mathbf{S}^\top = \hat{\Sigma} \mathbf{S} \hat{\Sigma}, \quad \hat{\Sigma} = \text{diag}(1, -1, 1, -1), \quad (8)$$

i.e. $\mathbf{R}_D^\top = \Sigma \mathbf{R}_D \Sigma$, $\mathbf{R}_U^\top = \Sigma \mathbf{R}_U \Sigma$ and $\mathbf{T}_U = \Sigma \mathbf{T}_D^\top \Sigma$ with $\Sigma = \text{diag}(1, -1)$, while the SH block is symmetric. Both hold to 50 digits at an interface, with and without attenuation, and plain $\mathbf{S} = \mathbf{S}^\top$ fails there by 0.31.

- **Unitarity** is sesquilinear and holds *only in the absence of loss*.

A lossless reciprocal system satisfies both. In the Kennett convention $\mathbf{S}^{-1} = \mathbf{S}^\dagger = \mathbf{S}^*$, which is the time-reversal statement that the conjugate outgoing field retraces the incident one (Waterman, 1976). Because the two properties constrain different combinations of entries, a numerical check should test both.

7 Unitarity of the Kennett recursion

The reflectivity recursion combines layer scattering matrices with the Redheffer star product (Redheffer, 1961; Kennett and Kerry, 1979). For two elements $\mathbf{S}^{(1)}$ above $\mathbf{S}^{(2)}$,

$$\mathbf{R}_D^{(12)} = \mathbf{R}_D^{(1)} + \mathbf{T}_U^{(1)} \mathbf{R}_D^{(2)} (\mathbf{I} - \mathbf{R}_U^{(1)} \mathbf{R}_D^{(2)})^{-1} \mathbf{T}_D^{(1)}, \quad (9)$$

together with the analogous expressions for $\mathbf{T}_D^{(12)}$, $\mathbf{R}_U^{(12)}$ and $\mathbf{T}_U^{(12)}$. The composite is the scattering matrix of the cascade. Since each element conserves flux, the cascade does too, so the star product of unitary matrices is unitary (Pozar, 2012). With modified coefficients a lossless stack therefore remains unitary at every step, and $\|\mathbf{S}^\dagger \mathbf{S} - \mathbf{I}\|$ directly measures accumulated numerical error. Displacement coefficients would require the flux weights of (4) to be carried through every step.

8 Numerical check

The interface coefficients were evaluated for $(\alpha, \beta, \rho)_1 = (3.0, 1.7, 2.3)$ and $(\alpha, \beta, \rho)_2 = (4.2, 2.4, 2.7)$, in km/s and g/cm³. Attenuation uses the complex slowness

$$s = \frac{1}{v} \frac{(2Q)^2 + i 2Q}{1 + (2Q)^2}, \quad \text{Im } s > 0, \quad (10)$$

with the branch $\text{Re } \eta > 0$, $\text{Im } \eta > 0$. The column $\|\cdot\|_{\max}$ reports the largest entry of the residual.

Two independent routes.

- **Python**, `kennett_layers.py`: the Aki-Richards closed forms with $\sqrt{\rho\eta}$ and $\sqrt{Z}$ normalisation, in double precision.

- **Mathematica, InterfaceUnitarity.wl**: continuity of the displacement–traction vector across $z = 0$ in the thesis energy-normalised eigenbasis D_z (Nestor 1996, §3.1), at 50 digits. With $\mathbf{b} = D_1(\mathbf{d}_1; \mathbf{u}_1) = D_2(\mathbf{d}_2; \mathbf{u}_2)$ and $Q = D_1^{-1}D_2$ in 3×3 blocks,

$$\mathbf{T}_D = Q_{11}^{-1}, \quad \mathbf{R}_D = Q_{21}\mathbf{T}_D, \quad \mathbf{R}_U = -\mathbf{T}_D Q_{12}, \quad \mathbf{T}_U = Q_{22} - Q_{21}\mathbf{T}_D Q_{12}. \quad (11)$$

The two share no code and no algebra. Both are flux-normalised but use different per-channel phase conventions, so they are compared through quantities invariant under unit-modulus rephasing: the magnitudes $|S_{ij}|$ and the eigenvalues of $\mathbf{S}^\dagger \mathbf{S}$. As a negative control the Python $\mathbf{S}$ is converted back to displacement-basis coefficients, $W^{-1/2} \mathbf{S} W^{1/2}$ with $W = \rho v^2 \eta$ from (3). The comparison must then fail, and it does, by 0.55 to 1.4 in every case.

Table 2. Unitarity residuals of the interface $\mathbf{S}$ from the Mathematica route, and its agreement with the Python route. The $p = 0.30$ rows exceed $1/\alpha_2 \approx 0.238$, so P in medium 2 is evanescent. “0” means zero to 50-digit working precision.

Case	Quantity	Value
Lossless, $p \in \{0, 0.1, 0.2\}$	$\ \mathbf{S}^\dagger \mathbf{S} - \mathbf{I}\ _{\max}$, P–SV and SH	0
Lossless, $p = 0.30$, all 4 channels	$\ \mathbf{S}^\dagger \mathbf{S} - \mathbf{I}\ _{\max}$, P–SV	1.56
Lossless, $p = 0.30$, 3 open channels	$\ \mathbf{S}_r^\dagger \mathbf{S}_r - \mathbf{I}\ _{\max}$	0
Lossless, $p = 0.30$, SH	$\ \mathbf{S}^\dagger \mathbf{S} - \mathbf{I}\ _{\max}$	0
$Q_P = 50$, $Q_S = 30$, $p = 0.1$	$\text{eig}(\mathbf{S}^\dagger \mathbf{S})$, P–SV	0.9892 to 1.0109
	$\text{eig}(\mathbf{S}^\dagger \mathbf{S})$, SH	0.99949, 1.00051
$Q_P = 20$, $Q_S = 10$, $p = 0.1$	$\text{eig}(\mathbf{S}^\dagger \mathbf{S})$, P–SV	0.9666 to 1.0346
	$\text{eig}(\mathbf{S}^\dagger \mathbf{S})$, SH	0.99848, 1.00152
All six cases	Python vs Mathematica, $ S_{ij} $, $\text{eig}(\mathbf{S}^\dagger \mathbf{S})$	$\leq 1.0 \times 10^{-15}$
All six cases	Displacement-basis control	≥ 0.55
Lossy, Kennett basis	$\ \mathbf{S} - \mathbf{S}^\dagger\ _{\max}$	0

The lossy rows show the point of §6. Eigenvalues of $\mathbf{S}^\dagger \mathbf{S}$ fall on *both* sides of unity, and the spread grows as Q falls, so the lossy interface matrix is not a contraction and (2) has no energy interpretation there. Reciprocity still holds exactly.

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